

ELYES THABET

Biomedical Engineering Student | Biosignal Processing & Wearable AI

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PROFILE

Fourth-year Biomedical Engineering student focused on physiological signal processing, multimodal biosignal learning, and wearable embedded systems. Developed a cervical-PPG silent-speech interface and co-authored a review in *Biosensors and Bioelectronics*. Research interests: cross-subject generalisation, data-efficient learning, and multimodal EMG-IMU decoding for assistive interfaces.

EDUCATION

Higher Institute of Medical Technologies of Tunis (ISTMT)

Tunis, Tunisia

National Engineering Diploma in Biomedical Engineering – Technological Innovation in Health 2022 – Present; expected 2027

- University of Tunis El Manar; ranked **3rd/40**; second engineering-year average **15.08/20** with Honors equivalent to a GPA of **3.5/4**. Selected grades: Signal Processing **18.00/20**, Embedded Systems **17.10/20**, Advanced Physiological Signal Processing **17.05/20**.

RESEARCH EXPERIENCE

Remote Research Intern – QUB Online Summer Research Programme

Jun. 22 – Jul. 31, 2026

Wellcome-Wolfson Institute for Experimental Medicine, Queen's University Belfast

Remote

- Conducting a supervised evidence synthesis on *GREM1* genetic variants, dysregulated expression, and associated phenotypes in human cancer within a research group led by Dr Derek Brazil.
- Completing journal clubs, research-paper analysis, scientific-writing and data-analysis training, weekly supervisor assignments, and a final group research project.

End-of-Year Research Project – Voice Assistance via Cervical PPG

Jan. – May 2026

ISTMT; supervised by Prof. Fatouma Boukadida

Tunis, Tunisia

- Built a neck-worn silent-speech system using MAX30102/ESP32 acquisition, filtering, segmentation, CWT time-frequency representations, and a dual-input CNN.
- Achieved only **64.3% mean K-fold accuracy** and **60.3% macro-F1** far above the probability chance 20% deployed live inference through an ESP32-to-Python pipeline and Flutter text-to-speech application **within only 3 months time**.

Research Summer Intern – International Internship Pilot Program

Jul. – Sep. 2025

National Chung Cheng University

Chiayi, Taiwan

- Reviewed more than 140 papers on transition-metal-dichalcogenide electrochemical biosensors and contributed to the evidence synthesis and conceptual framework of a published review article.
- Developed a five-axis clinical-translation framework covering performance, stability, reproducibility, bio-interface compatibility, and system readiness.

Research Intern – Multimodal Biomedical AI

Jul. – Sep. 2024

AI4U Research Unit

Tunis, Tunisia

- Built an exploratory multimodal pipeline combining text, image, and 16-channel EEG features; used SHAP and attention visualisation to inspect model behaviour.

PUBLICATION

Chen, T.-H.; Koh, K.-Y.; Mukundan, A.; Karmakar, R.; **Thabet, E.**; et al.

2026

- From surface to signal: Clinical pathways for transition metal dichalcogenide-based electrochemical biosensors. Biosensors and Bioelectronics*, 311, 118913. DOI: 10.1016/j.bios.2026.118913.

SELECTED ENGINEERING PROJECTS

Elderly Fall Detection System | STM32F407, ESP32, IMU, Embedded C, Flutter

2024

- Developed interrupt-driven IMU acquisition, multi-axis sensor fusion, and digital filtering for real-time movement classification; relayed alerts through ESP32 to a Flutter dashboard.

TECHNICAL SKILLS

- Biosignal Processing:** PPG, EEG, EMG, ECG and IMU data; digital filtering, FFT, CWT/time-frequency analysis, segmentation, normalisation, signal-quality checks; MATLAB and Python.
- Machine Learning:** PyTorch, TensorFlow, scikit-learn, CNNs, multimodal fusion, model evaluation, SHAP and attention-based interpretation.
- Embedded and Research Methods:** STM32F4, ESP32, Embedded C/C++, I2C/SPI/UART, interrupt-driven acquisition; literature review, evidence synthesis, scientific writing and journal-club presentation. Languages: Arabic (native), English (C1; EnglishScore 560), French (fluent).